



14.1

Determinant (5)

14.1.1 Determinant: ISRO CSE 2008 | Question: 31



If the two matrices $\begin{bmatrix} 1 & 0 & x \\ 0 & x & 1 \\ 0 & 1 & x \end{bmatrix}$ and $\begin{bmatrix} x & 1 & 0 \\ x & 0 & 1 \\ 0 & x & 1 \end{bmatrix}$ have the same determinant, then the value of x is

- A. $\frac{1}{2}$ B. $\sqrt{2}$ C. $\pm \frac{1}{2}$ D. $\pm \frac{1}{\sqrt{2}}$

isro2008 linear-algebra matrix determinant

Answer key

14.1.2 Determinant: ISRO CSE 2009 | Question: 60



If two adjacent rows of a determinant are interchanged, the value of the determinant

- A. becomes zero B. remains unaltered
C. becomes infinitive D. becomes negative of its original value

isro2009 linear-algebra matrix determinant

Answer key

14.1.3 Determinant: ISRO CSE 2009 | Question: 61



If $\begin{vmatrix} 3 & 3 \\ x & 5 \end{vmatrix} = 3$ then the value of x is

- A. 2 B. 3 C. 4 D. 5

isro2009 linear-algebra matrix determinant

Answer key

14.1.4 Determinant: ISRO CSE 2009 | Question: 63



$$\begin{vmatrix} 265 & 240 & 219 \\ 240 & 225 & 198 \\ 219 & 198 & 181 \end{vmatrix} =$$

- A. 779 B. 679 C. 0 D. 256

isro2009 linear-algebra matrix determinant

Answer key

14.1.5 Determinant: ISRO CSE 2023 | Question: 30



Given P is a matrix $= \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 0 & 3 & 1 \\ 6 & 4 & 2 & 8 \\ 2 & 5 & 1 & 3 \end{pmatrix}$

If $\det |P|$ denotes the determinant of matrix P , then which of the following is true:

- A. $\det |P|$ is indeterminate
 C. $\det |P| = 0$

- B. $\det |P|$ is negative
 D. None of the above

isro-cse-2023 linear-algebra determinant matrix

Answer key 

14.2 Eigen Value (2)

14.2.1 Eigen Value: ISRO CSE 2007 | Question: 09



Eigen vectors of $\begin{bmatrix} 1 & \cos \theta \\ \cos \theta & 1 \end{bmatrix}$ are

- A. $\begin{bmatrix} a^n & 1 \\ 0 & a^n \end{bmatrix}$
 B. $\begin{bmatrix} a^n & n \\ 0 & a^n \end{bmatrix}$
 C. $\begin{bmatrix} a^n & na^{n-1} \\ 0 & a^n \end{bmatrix}$
 D. $\begin{bmatrix} a^n & na^{n-1} \\ -n & a^n \end{bmatrix}$

isro2007 linear-algebra matrix eigen-value

Answer key 

14.2.2 Eigen Value: ISRO-2013-33



What is the matrix transformation which takes the independent vectors $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and transforms them to $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ respectively?

- A. $\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$
 B. $\begin{pmatrix} 0 & 0 \\ 0.5 & 0.5 \end{pmatrix}$
 C. $\begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$
 D. $\begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$

isro2013 linear-algebra eigen-value

Answer key 

14.3 Matrix (6)

14.3.1 Matrix: ISRO CSE 2008 | Question: 34



If a square matrix A satisfies $A^T A = I$, then the matrix A is

- A. Idempotent B. Symmetric C. Orthogonal D. Hermitian

isro2008 linear-algebra matrix

Answer key 

14.3.2 Matrix: ISRO CSE 2009 | Question: 59



A square matrix A is called orthogonal if $A'A =$

- A. I B. A C. $-A$ D. $-I$

isro2009 linear-algebra matrix

Answer key

14.3.3 Matrix: ISRO CSE 2009 | Question: 62



If A, B, C are any three matrices, then $A' + B' + C'$ is equal to

- A. a null matrix B. $A + B + C$ C. $(A + B + C)'$ D. $-(A + B + C)$

isro2009 linear-algebra matrix

Answer key

14.3.4 Matrix: ISRO CSE 2011 | Question: 36



If A and B are square matrices with same order and A is symmetric, then $B^T A B$ is

- A. Skew symmetric
B. Symmetric
C. Orthogonal
D. Idempotent

isro2011 linear-algebra matrix

Answer key

14.3.5 Matrix: ISRO CSE 2017 | Question: 1



If A is a skew symmetric matrix then A^t is

- A. Diagonal matrix B. A C. 0 D. $-A$

isro2017 linear-algebra matrix

Answer key

14.3.6 Matrix: ISRO-DEC2017-17



If C is a skew-symmetric matrix of order n and X is $n \times 1$ column matrix, then $X^T C X$ is a

- A. scalar matrix B. null matrix C. unit matrix D. matrix will all elements 1

isrodec2017 matrix linear-algebra

Answer key

14.4

Rank of Matrix (1)

14.4.1 Rank of Matrix: ISRO CSE 2014 | Question: 72



The rank of the matrix $A = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 9 & 5 & 2 & 2 \\ 7 & 1 & 0 & 4 \end{pmatrix}$ is _____.

- A. 0 B. 1 C. 2 D. 3

Answer key 

14.5

Vector Space (1)

14.5.1 Vector Space: ISRO-DEC2017-10



If vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular to each other, then value of λ is

A. $\frac{2}{5}$

B. 2

C. 3

D. $\frac{5}{2}$

Answer key 

Answer Keys

14.1.1	A
14.2.1	Q-Q
14.3.4	B

14.1.2	D
14.2.2	D
14.3.5	D

14.1.3	C
14.3.1	C
14.3.6	B

14.1.4	C
14.3.2	A
14.4.1	C

14.1.5	Q-Q
14.3.3	C
14.5.1	D